

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

INDUSTRIAL INTERNSHIP REPORT

ON

AGRICULTURE CROP PRODUCTION PREDICTION IN INDIA

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Internship Duration: 5 Weeks

Internship facilitated by upSkill Campus in collaboration with
UniConverge Technologies Pvt. Ltd. (UCT)

Academic Course: BCA DATA SCIENCE HONOURS

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Executive Summary

This report describes the six-week industrial internship facilitated by upSkill Campus in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). The project, Agriculture Crop Production Prediction in India, applies a supervised machine-learning approach to historical agricultural production data.

The work included inspecting the available CSV datasets, preparing a cleaned dataset, organizing the data for modeling, training a Random Forest Regressor, evaluating its predictions, and creating a visualization comparing actual and predicted values. The project was implemented in Python with common data-analysis and machine-learning tools.

The recorded evaluation results were MAE = 29.7601, RMSE = 38.7184, and $R^2 = 0.5661$. These values are reported as obtained in the project; interpretation is limited by the dataset, feature selection, and evaluation setup. The project provided practical experience in data preparation, model development, evaluation, visualization, and version control.

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1. Preface

This report summarizes the work completed during a six-week internship program facilitated by upSkill Campus with UniConverge Technologies Pvt. Ltd. The internship offered an opportunity to apply classroom learning to a practical data-science project.

The project focused on Agriculture Crop Production Prediction in India. The work progressed through dataset inspection, cleaning, preparation, model training, evaluation, and documentation. Completing the project helped develop a structured approach to solving a problem using data and programming.

I am thankful to upSkill Campus, UniConverge Technologies Pvt. Ltd., The IoT Academy, and the mentors and instructors who supported the internship. I also thank SRM Institute of Science and Technology for supporting my learning. This experience reinforced the importance of consistent practice, careful validation, and clear communication.

Message to peers: approach each project step by step, keep your files organized, record results accurately, and ask for help whenever a technical concept is unclear.

2. Introduction

Agriculture is an important sector in India, and crop production data can be studied to understand historical patterns and support planning. Machine learning provides methods for learning relationships from recorded data. In this project, a regression model was trained to estimate crop production from the available prepared dataset.

2.1 About UniConverge Technologies Pvt. Ltd. (UCT)

The supplied internship template describes UniConverge Technologies Pvt. Ltd. as a company established in 2013 that works in digital transformation and industrial solutions. It identifies areas including Internet of Things (IoT), cybersecurity, cloud computing, machine learning, communication technologies, Java full stack, Python, and front-end development.

2.2 About upSkill Campus and The IoT Academy

According to the supplied template, upSkill Campus, The IoT Academy, and UCT facilitated the internship process. The IoT Academy is described as the EdTech division of UCT and as a provider of executive certification programs in collaboration with academic institutions.

2.3 Internship Objectives

- Gain practical experience with an industry-oriented project.
- Apply Python and machine-learning concepts to a dataset.
- Practice data inspection, cleaning, preparation, and visualization.
- Train and evaluate a regression model using measurable metrics.
- Improve documentation, version-control, and problem-solving skills.

2.4 References

1. Internship report template supplied by the learner (Sample_InternshipReport_USC_UCT.docx).
2. Learner's Agriculture Crop Production Prediction project repository:
https://github.com/Radha231007/Agriculture_Crop_Production_Prediction
3. Project dataset files and outputs maintained in the learner's project repository.

2.5 Glossary

Term	Meaning
ML	Machine Learning
CSV	Comma-Separated Values
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
R ²	Coefficient of determination
Model	A learned mapping used to make predictions
Random Forest	An ensemble learning algorithm based on multiple decision trees

3. Problem Statement

The project addresses the task of using historical agriculture crop-production records to build a machine-learning model that estimates production from the available input features. The practical challenge is to prepare inconsistent or separately stored data into a usable form and evaluate whether a trained model can produce reasonable estimates.

Project objective: develop and evaluate a Random Forest regression model for agriculture crop production using the dataset provided for the project.

Scope note: this is a learning project based on the available dataset. The output should not be treated as an official agricultural forecast or used alone for operational decisions.

4. Existing and Proposed Solution

4.1 Existing Approach

A basic alternative is to inspect historical production values manually or use simple averages and trends. Such approaches can be limited when several input variables interact. The project explores a data-driven regression model as an additional analytical approach.

4.2 Proposed Solution

The proposed workflow prepares the available records, selects the model inputs and target, trains a Random Forest Regressor, evaluates predictions using MAE, RMSE, and R^2 , and saves a comparison plot. The workflow provides a repeatable starting point for further experimentation.

4.3 Code Submission (GitHub)

Project repository: https://github.com/Radha231007/Agriculture_Crop_Production_Prediction

4.4 Report Submission (GitHub)

Report PDF link: To be added after the final PDF is uploaded to the required submission repository.

5. Proposed Design / Model

5.1 High-Level Workflow

1. Input: agriculture CSV data files.
2. Inspect columns, dimensions, and missing values.
3. Clean and reshape the relevant production data into a structured dataset.
4. Prepare model features and target variable.
5. Split data into training and testing portions.
6. Train the Random Forest Regressor.
7. Evaluate using MAE, RMSE, and R^2 .
8. Save model output and an actual-versus-predicted visualization.

5.2 Implementation Details

The project was developed in Python in a local project environment. The prepared dataset was saved as `data/processed/agriculture_cleaned.csv`. The trained model was saved as `models/agriculture_production_random_forest.pkl`. The evaluation visualization was saved as `results/figures/actual_vs_predicted.png`.

The recorded model was a Random Forest Regressor. The project used Python data-science libraries, including pandas, scikit-learn, matplotlib, and seaborn as part of the workflow.

5.3 Interfaces and Outputs

- Cleaned dataset: `data/processed/agriculture_cleaned.csv`
- Saved model: `models/agriculture_production_random_forest.pkl`
- Evaluation plot: `results/figures/actual_vs_predicted.png`
- Source code and project documentation: GitHub repository

Insert screenshots of the project folder, model evaluation output, and actual-versus-predicted plot here if required by the college.

6. Performance Test

The model was evaluated using three regression metrics. MAE measures average absolute prediction error; RMSE gives greater weight to larger errors; R^2 describes the proportion of target variation explained by the model under the evaluation setup.

6.1 Test Plan / Test Cases

- Check that the selected CSV data can be read.
- Inspect missing values and prepare the cleaned dataset.
- Confirm that the model training process completes.
- Generate predictions for the held-out evaluation data.
- Calculate MAE, RMSE, and R^2 .
- Confirm that the trained model and plot are saved to the project folders.

6.2 Test Procedure

The prepared agriculture dataset was used for model development. A Random Forest Regressor was trained and evaluated using the project's train/test procedure. The resulting metrics were recorded, and an actual-versus-predicted plot was generated. Detailed split settings and feature-level experiment logs should be checked against the source scripts if a formal reproducibility audit is required.

6.3 Performance Outcome

Metric	Recorded result
Mean Absolute Error (MAE)	29.7601
Root Mean Squared Error (RMSE)	38.7184
R^2 Score	0.5661

The recorded R^2 score indicates that the model explains a portion of the variation in the evaluation target, while the error metrics show that prediction errors remain. Performance may be improved through additional data validation, feature engineering, model comparison, and careful evaluation.

7. My Learnings

- Learned to inspect and organize multiple CSV files.
- Practiced data cleaning and preparation using Python.
- Understood the basic workflow of supervised regression.
- Implemented and evaluated a Random Forest Regressor.
- Learned the meaning and use of MAE, RMSE, and R^2 .
- Created a visualization to compare actual and predicted values.
- Practiced saving project artifacts and maintaining source code on GitHub.
- Improved debugging, documentation, and step-by-step problem solving.

The internship helped connect programming and machine-learning concepts with a practical dataset. These skills can support future study and project work in data analytics and machine learning.

8. Future Work Scope

- Verify dataset definitions, units, and year/state/crop coverage.
- Add suitable explanatory features where reliable data is available.
- Compare Random Forest with baseline, Decision Tree, and other regression models.
- Use cross-validation or time-aware evaluation where appropriate.
- Tune model parameters and report uncertainty alongside point predictions.
- Build a simple interface to select crop and relevant inputs and display model output.
- Document dataset provenance, limitations, and reproducible setup instructions.

The current project is a learning prototype. Any future deployment would require domain review, more extensive validation, and clear communication of uncertainty and limitations.

9. LINKS

1. Project / Code GitHub link : https://github.com/Radha231007/Agriculture_Crop_Production_Prediction